De La Salle University

Fama-French 5-Factor Model Evaluation

A Financial Analytics Project
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1 Introduction

This document provides a rigorous evaluation of the **Fama-French 5-Factor Model** using R. The model extends the Capital Asset Pricing Model (CAPM) by incorporating Size, Value, Profitability, and Investment factors alongside standard market risk (Fama & French, 1993, 2015). The core objective is to determine the efficacy of these five factors in explaining the excess returns of a portfolio comprising eight selected assets.

/subsectionTransition from CAPM to Fama-French 5-Factor Model The Fama-French Model, developed by Eugene F. Fama and Kenneth R. French in September 1992 in their paper titled 'The Cross-Section of Expected Stock Return', was originally composed of three (3) factors and serves as an augmentation to the Capital Asset Pricing Model (CAPM). The primary purpose of the introduction of this model was to demonstrate that market risk alone cannot fully explain stock returns. Thus, the transition from a single-factor (beta) linear model (CAPM) to a multi-factor linear model led to the creation of a model that could provide a multidimensional assessment for expected returns. Here, the CAPM model was expanded to include firm-specific variables that are historically known to have an effect on returns.

The initial Fama-French Model was composed of the following systematic risk factors. First, Market Risk (Rf), taken from the original CAPM model, represents the excess returns of the overall market over the overall risk-free rate. Next, SMB (small minus big) represents the idea that small-cap stocks tend to outperform large-cap stocks over time. Lastly, there is HML (high minus low) or the value factor; stocks with higher BM ratios tend to outperform stocks with low BM ratios. The three-factor model was originally developed to encapsulate the relationship between average return and price ratios (such as book-to-market ratio) (Fama and French, 2015).

Moving forward, the Fama-French 3-Factor Model was further augmented by Fama and French in 2015 in their working paper titled 'A five-factor asset pricing model'. This model is composed of the three factors mentioned above with the further additions of RMW (robust minus weak) and CMA (conservative minus aggressive). The factor of RMW represents the difference between the returns of stocks with robust and weak profitability. Meanwhile, CMA represents the difference between the returns on diversified portfolios of stocks of low and high investment firms, which are referred to as conservative and aggressive (Fama and French, 2015). These labels refer to a company's investment policy, whether it can be considered conservative or aggressive (Mosoeu and Kodongo, 2022). This new model with these two newly added variables is what is now referred to as the Fama-French 5-Factor Model.

1.1 Objectives

- **Empirical Analysis:** Assess how well the five factors (Market, SMB, HML, RMW, CMA) explain asset returns.
- **Statistical Interpretation:** Analyze alpha and beta coefficients for statistical signif-

icance and economic meaning.

2 Methodology

The methodology involves a systematic technical workflow utilizing the Kenneth French data library (French, 2024) and R-based statistical modeling.

2.1 Asset Selection

The following 8 assets were selected representing tilts toward Market, SMB, HML, RMW, and CMA. Each selection is justified based on historical factor exposure.

2.2 Descriptive Statistics

Historical return data for the selected assets is analyzed using mean, median, standard deviation, skewness, and kurtosis to understand the distribution and risk profile of the portfolio.

3 Results

[Detailed results, regression coefficients, and visual analysis to be imported from Calculations.Rmd]

4 Conclusion

The study concludes with an academic verdict on the model's efficacy for the selected portfolio, linking empirical findings to practical investment implications.

References

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